

ARM Task Implementation

Name: Srinidhi A
ID: COMETFWC053

Question 36 (GATE EC 2009)

If $X = 1$ in the logic equation

$$F = [X + Z(Y + (Z + XY)')][(\bar{Y} + \bar{Z}(X + Y))]$$

and $F = 1$, then the correct option is:

- (a) $Y = Z$
- (b) $Y = \bar{Z}$
- (c) $Z = 1$
- (d) $Z = 0$

Detailed Solution

Given:

$$F = [X + Z(Y + (Z + XY)')][(\bar{Y} + \bar{Z}(X + Y))]$$

Substitute $X = 1$:

$$F = [1 + Z(Y + (Z + Y)')][(\bar{Y} + \bar{Z}(1 + Y))]$$

Using Boolean identities:

$$1 + A = 1, \quad 1 + Y = 1$$

Thus,

$$F = \bar{Y} + \bar{Z}$$

Using De Morgan's theorem:

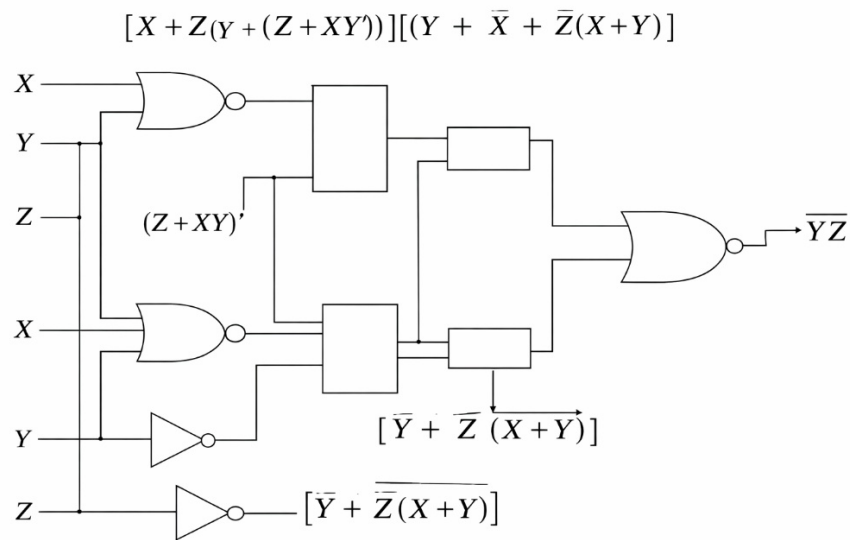
$$F = \bar{Y}\bar{Z}$$

Correct Option: (a) $Y = Z$

Truth Table

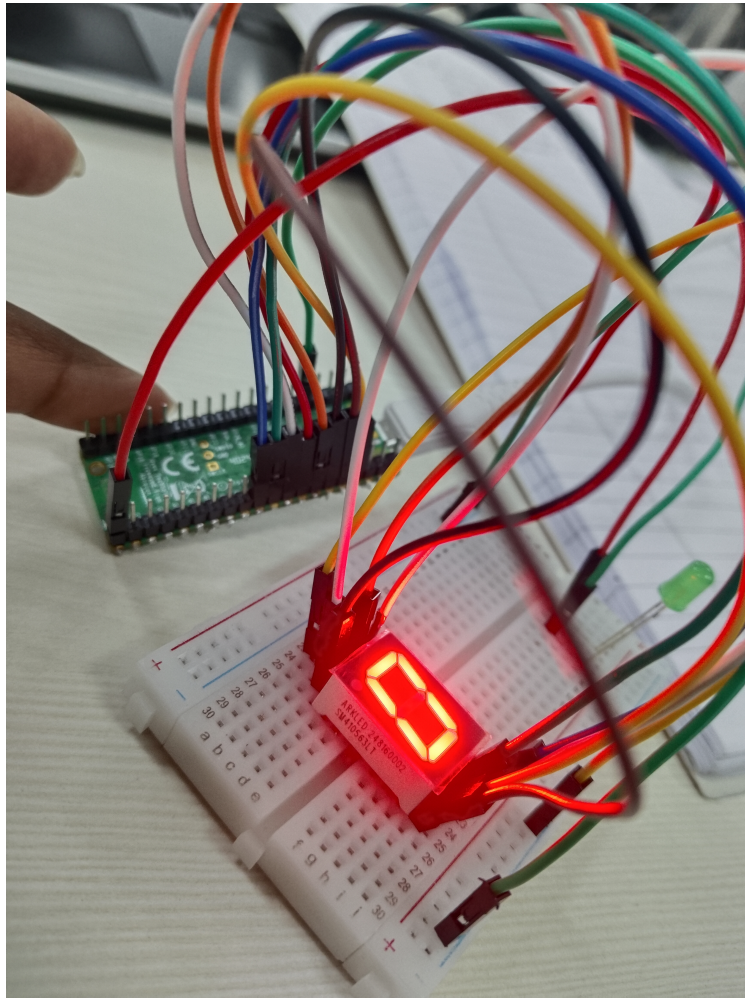
Y	Z	$F = \overline{Y}Z$
0	0	1
0	1	1
1	0	1
1	1	0

Gate Diagram Representation



Simplified implementation: 2-input NAND Gate.

Hardware Implementation



Steps to Perform the Task

1. Connect switch Y to GPIO14 (pull-up).
2. Connect switch Z to GPIO15 (pull-up).
3. Connect LED to GPIO16 via 220Ω resistor.
4. Connect 7-segment display to GPIO pins 2–8.
5. Upload MicroPython code.
6. Toggle Y and Z switches and observe output.
7. Verify output follows NAND logic.

Conclusion

After simplification, the Boolean expression reduces to:

$$F = \overline{Y} \overline{Z}$$